Assignment_01, May.2014

Course “Advanced Networking”

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Goal:

- Learn about IP address
- Learn how end host routing works and basic commands for routing table, arp cache table, and network interface
- Learn how to interpret the IP classless notation and routing table entries
- Learn how to partition a subnet and assign gateway address

Policy:

- No copy, No Cheating. If you violate this policy, ZERO scores for BOTH: copy & giving copy!
- Save your assignment solution into a PDF file with format:
  - Yourname_YourStudentID_assign01.PDF
- Send your assignment to: advanced.networking.submission@gmail.com
- Deadline: 05.00pm, Sat 26/04/2014!

Problem 01: *(15 marks)* IP Address and Routing

A router has the following (CIDR) entries in its routing table:

Address/mask	Next hop
192.168.56.0/22	Interface 0
192.168.60.0/22	Interface 1
192.53.40.0/25	Router 1
default	Router 2

For each of the following IP addresses, what does the router do if a packet with that address arrives?
(a) 192.168.63.10
(b) 192.168.57.14
(c) 192.168.52.2
(d) 192.53.40.7
(e) 192.53.56.7

Problem 2: *(15 marks)* IP Address and Routing

Given the following routing table on dragon.mit.edu.
dragon.mit.edu> netstat -rn

```
Kernel IP routing table
Destination    Gateway        Genmask        Flags MSS Window irtt Iface
192.168.160.0  0.0.0.0        255.255.224.0  U     0   0      0    eth0
192.168.80.0   0.0.0.0        255.255.255.0  U     0   0      0    wlan0
169.254.0.0    0.0.0.0        255.255.0.0    U     0   0      0    eth0
127.0.0.0      0.0.0.0        255.0.0.0      U     0   0      0    lo
0.0.0.0        192.168.160.1  0.0.0.0        UG    0   0      0    eth0
```

From the first entry of the routing table, it indicates dragon located in a wired Ethernet subnet with IP address 192.168.160.0 and genmask (or called netmask) 255.255.224.0. The second entry indicates arbor also has wireless LAN card on a subnet identified with 192.168.80.0 and genmask 255.255.255.0.

What is the equivalent Classless IP addess notation for the wired Ethernet subnet?
What is the equivalent Classless IP address notation for the wireless subnet? *(5 marks)*

How many IP addresses can be assigned to machines in this wireless subnet (assume one is used by the router for the grateway interface, one used for broadcast, and one used for identifying the subnet itself)? *(5 marks)*

What command(s) you use to find out the MAC address of the gateway interface for 192.168.160.0 subnet? *(5 marks)*

Problem 3: *(10 marks)* Assign IP addresses to computer labs
Assume you are given the assignment of setting three different computer labs. Each lab is configured with a file server, a printer and 30, 32, and 60 PCs respectively. You talked to the network administrator and was given 192.168.63.0/24 subnet for these three labs.

What are the network address, the broadcast address, and the gateway ip addresses you would like to assign to each of the three subnets? *(5 marks)*

Can 192.168.63.0/27 be enough for the first lab? Why is that? *(5 marks)*

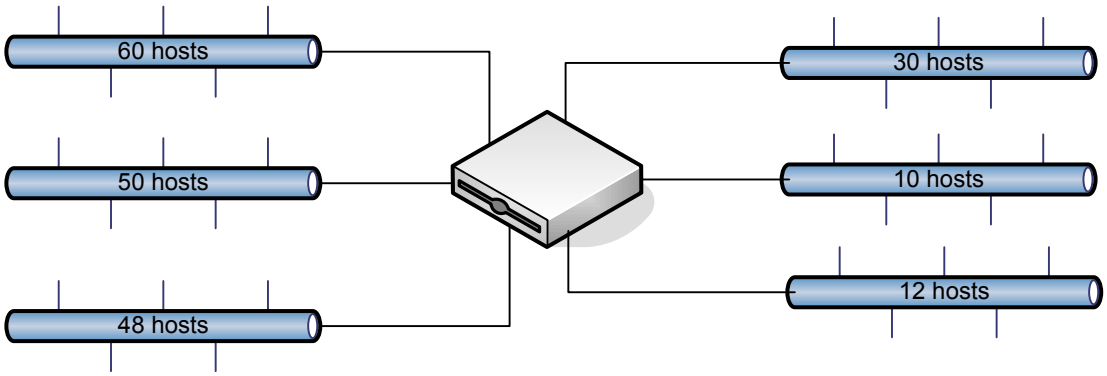
Problem 4: *(10 marks)* IP Addresses

For the following IPv4 subnets, indicate the smallest and the largest IPv4 address inside the subnet:

- 8.0.0.0/8
- 172.12.0.0/16
- 200.123.42.128/25
- 12.1.2.0/13

Problem 5: *(20 marks)* IP Subnet

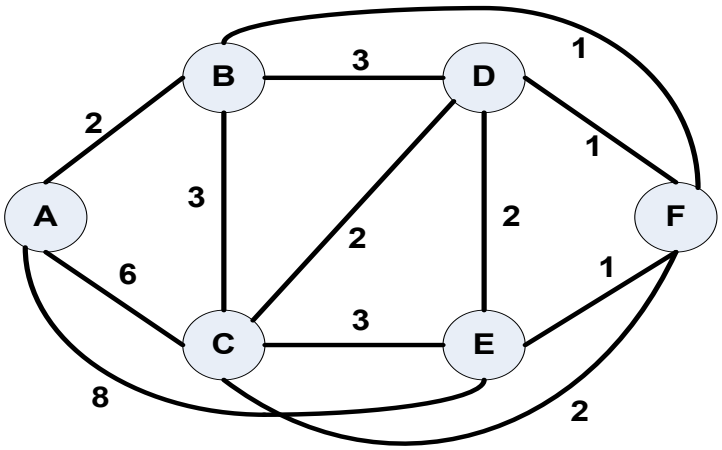
Given the network address 192.168.10.0/24, and the following network topology:



Propose a subnet solution to allocate the network address range: 192.169.10.0/24 into 6 network segments in the above figure.

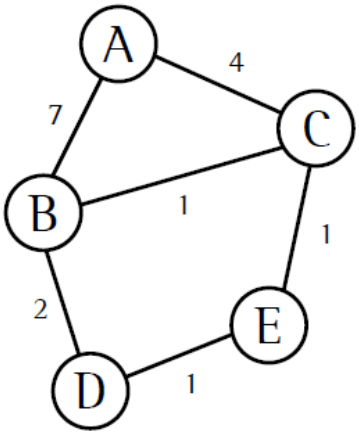
Problem 6: *(20 marks)* IP Routing Algorithm

Find the shortest paths from the source node A to the other nodes using Dijkstra's algorithm. Show all steps towards your solution.



<u>Problem 7:</u> *(10 marks)* Distance Vector Routing

Given the below network topology (the number represents the link cost):



What is the state of the routing table for each node before any distance vectors were exchanged?
(5 marks)

Now, exchange a few distance vectors between the routers and again, determine the routing tables which have changed.
(5 marks)

<u>Notes:</u> the state of the routing table for each node should be shown in the below form:
Router X

dest	cost	via node